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E500 Organic Elemental Analyzer

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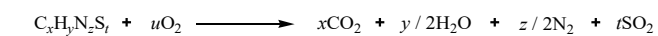
Basic Principle

Organic elements primarily refer to the five elements carbon (C), hydrogen (H), nitrogen (N), sulfur (S), and oxygen (O), which are the most abundant and common elements found in organic compounds.

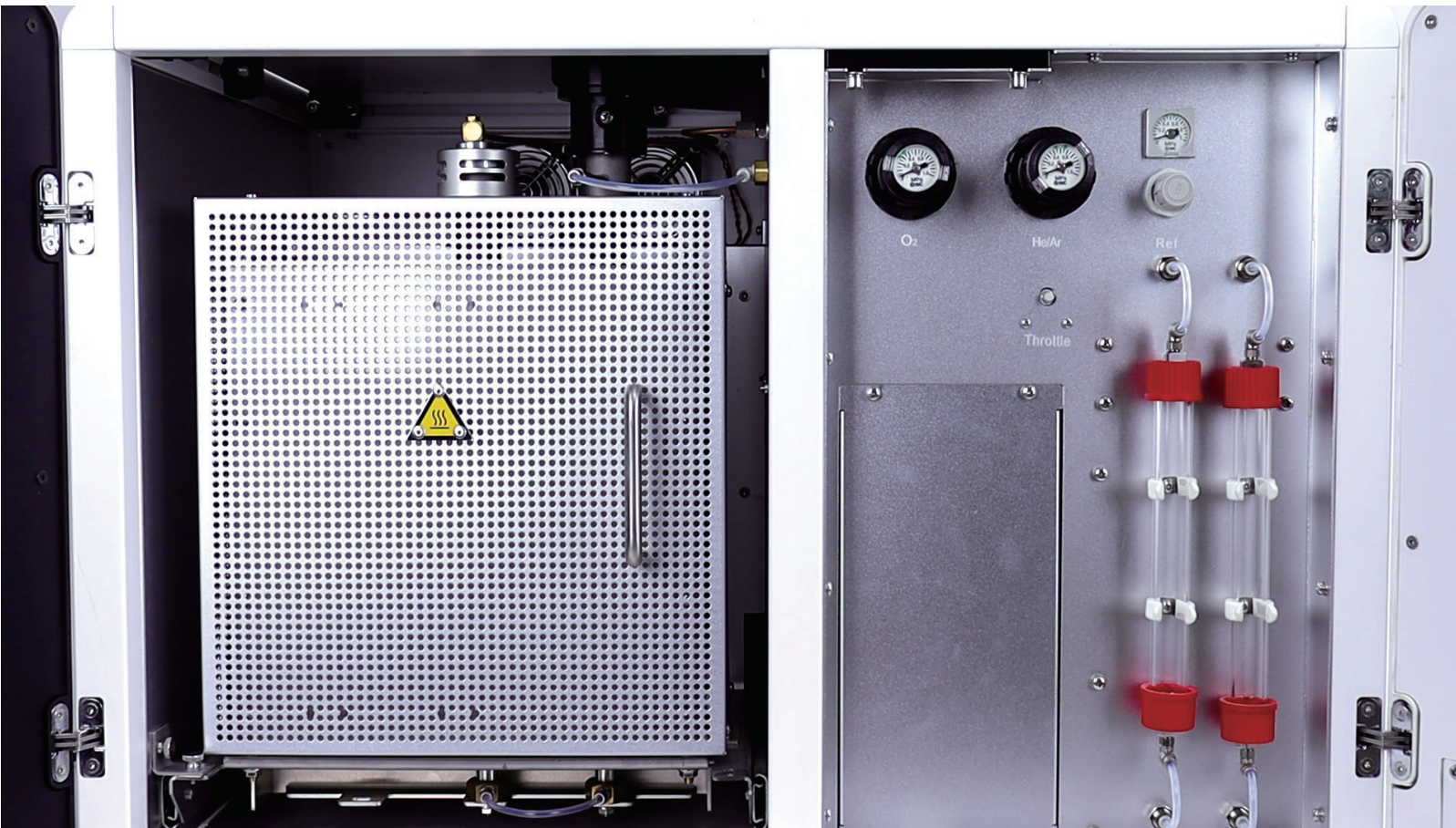
By determining the content of these organic elements within a compound, the proportional composition of each element can be established, leading to the derivation of the compound's empirical formula.

Modern Organic Elemental Analyzers operate based on principles developed from Fritz Pregl's method for measuring carbon and hydrogen and Jean-Baptiste Dumas's method for measuring nitrogen. They utilize the principle of high-temperature combustion analysis to determine the content of conventional organic elements in samples.

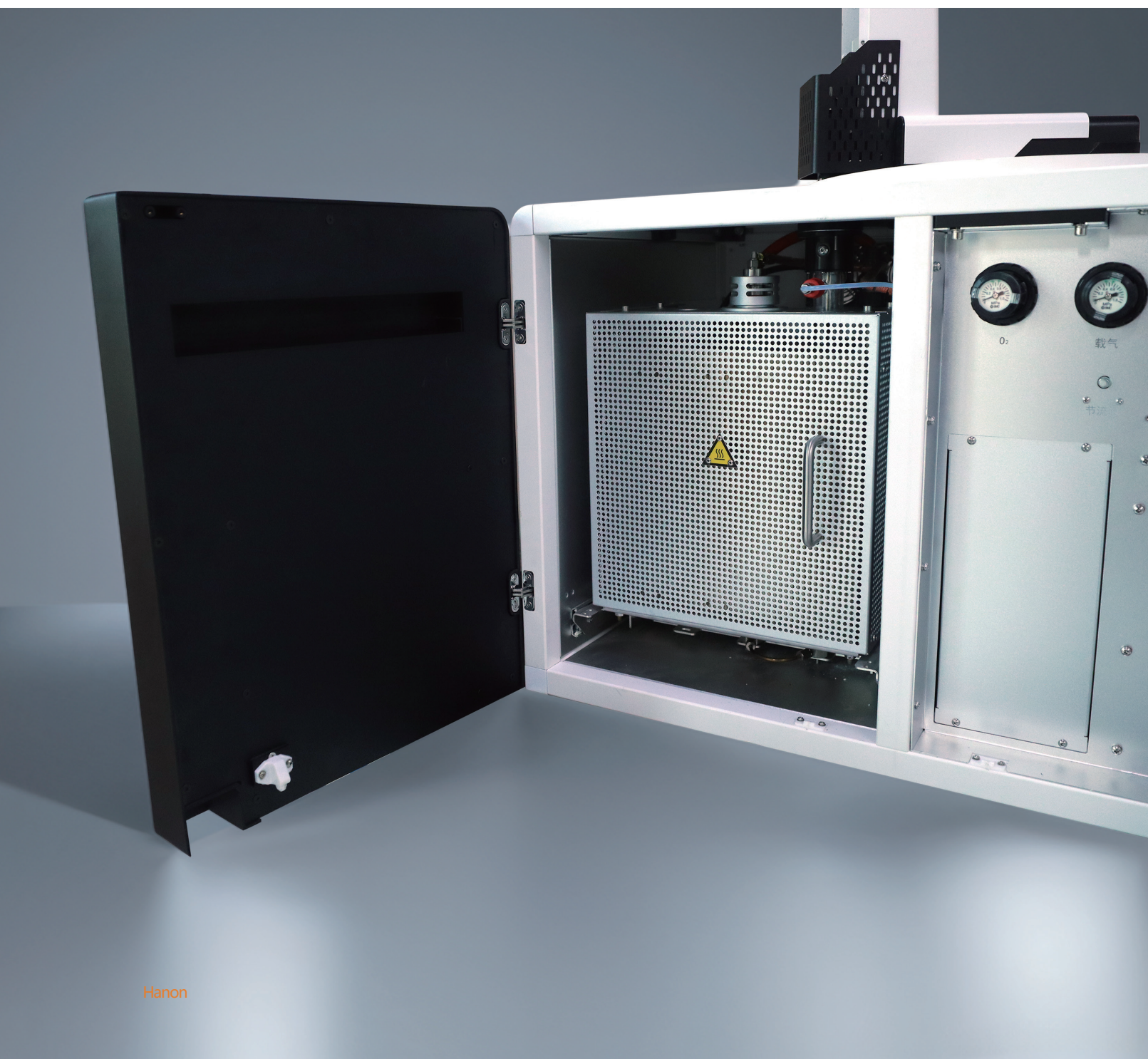
The main elements in organic matter are carbon (C), hydrogen (H), nitrogen (N), sulfur (S), and oxygen (O). Under high-temperature, oxygen-rich conditions, organic matter combusts. During combustion, these organic elements are converted into their respective stable forms, such as CO₂, H₂O, N₂, SO₂, etc. The carrier gas transports these combustion product gases through adsorption columns for separation via adsorption-desorption. The separated components then sequentially enter the Thermal Conductivity Detector (TCD) for content detection.



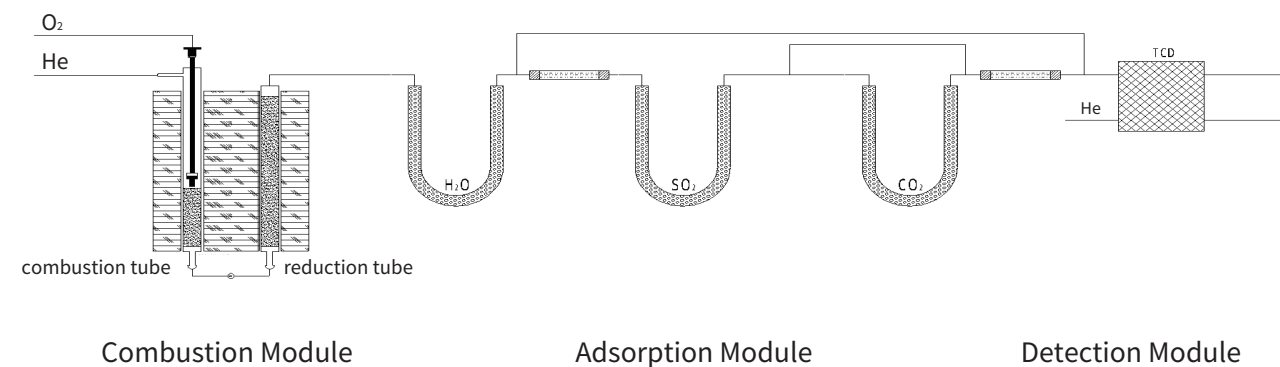
For oxygen (O) determination, the sample is pyrolyzed within a high-temperature pyrolysis tube. The pyrolysis products are carried by the carrier gas through activated carbon packing, converting all oxygen into carbon monoxide (CO). The CO then enters the detector. Quantitative detection of CO allows the calculation of the oxygen content in the sample.



Hanon Technology is a scientific instrument service provider focused on analytical instruments and method research, with years of accumulated expertise in organic elemental analysis technology. In 2015, Hanon pioneered the launch of the first Dumas nitrogen analyzer in China. Building on years of experience in developing, manufacturing, and applying combustion-based nitrogen analyzers, Hanon has now introduced the brand-new C/H/N/S/O five-element Organic Elemental Analyzer.



Operating Procedure ----- C/H/N/S



Sample Introduction: Samples (powder/liquid) are wrapped in tin foil/sn capsules, precisely weighed, placed into ceramic sample crucibles, and arranged in sequence within the autosampler sample tray.

A high-temperature alloy sample probe delivers the crucible and sample into the combustion tube within the furnace. The entire introduction process occurs under a purge of carrier gas, ensuring ambient air does not enter the combustion tube.

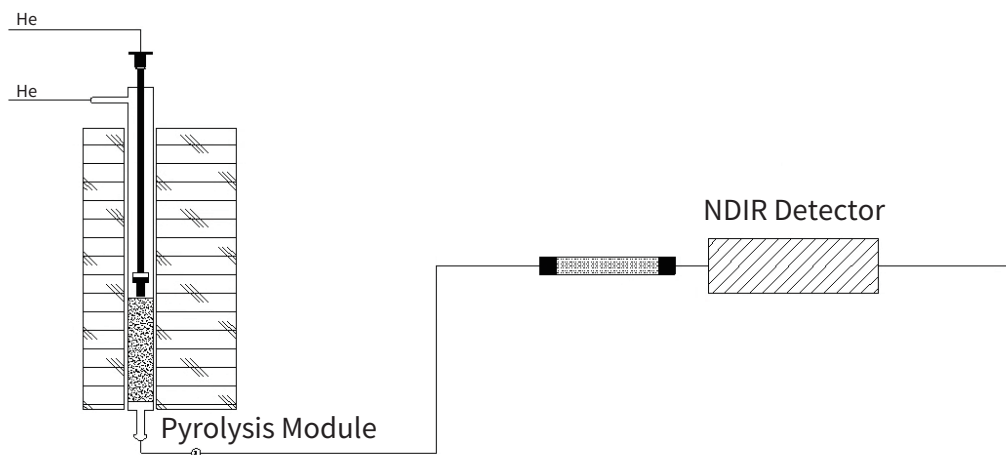
Combustion - Reduction: The ceramic furnace heats the combustion tube to 1150°C. An oxygen lance precisely directs combustion oxygen to the sample position. The gaseous products from complete combustion are carried by the carrier gas into the reduction tube located within the furnace.

Inside the reduction tube, catalysts/reducing agents convert the nitrogen oxides (NO_x) produced during combustion into N₂. The N₂, along with CO₂, H₂O, and SO₂, is carried by the carrier gas into the adsorption-separation unit.

Adsorption - Separation: This unit consists of three adsorption columns filled with specific molecular sieve adsorbent materials. These columns selectively adsorb H₂O, SO₂, and CO₂ respectively. N₂ passes through unadsorbed and is the first to enter the TCD for detection. Subsequently, CO₂, H₂O, and SO₂ are sequentially desorbed in that order, and each gas component enters the TCD detector one after another.

Detection: A high-throughput TCD thermal conductivity detector, featuring a reference-measurement dual-path design, converts the concentration information of the target gases in the measurement path into electrical signals. Optimized algorithms, combined with sample weight data and corresponding calibration curves, allow the software workstation to calculate the content of the target elements in the sample.

Operating Procedure ----- Oxygen Mode



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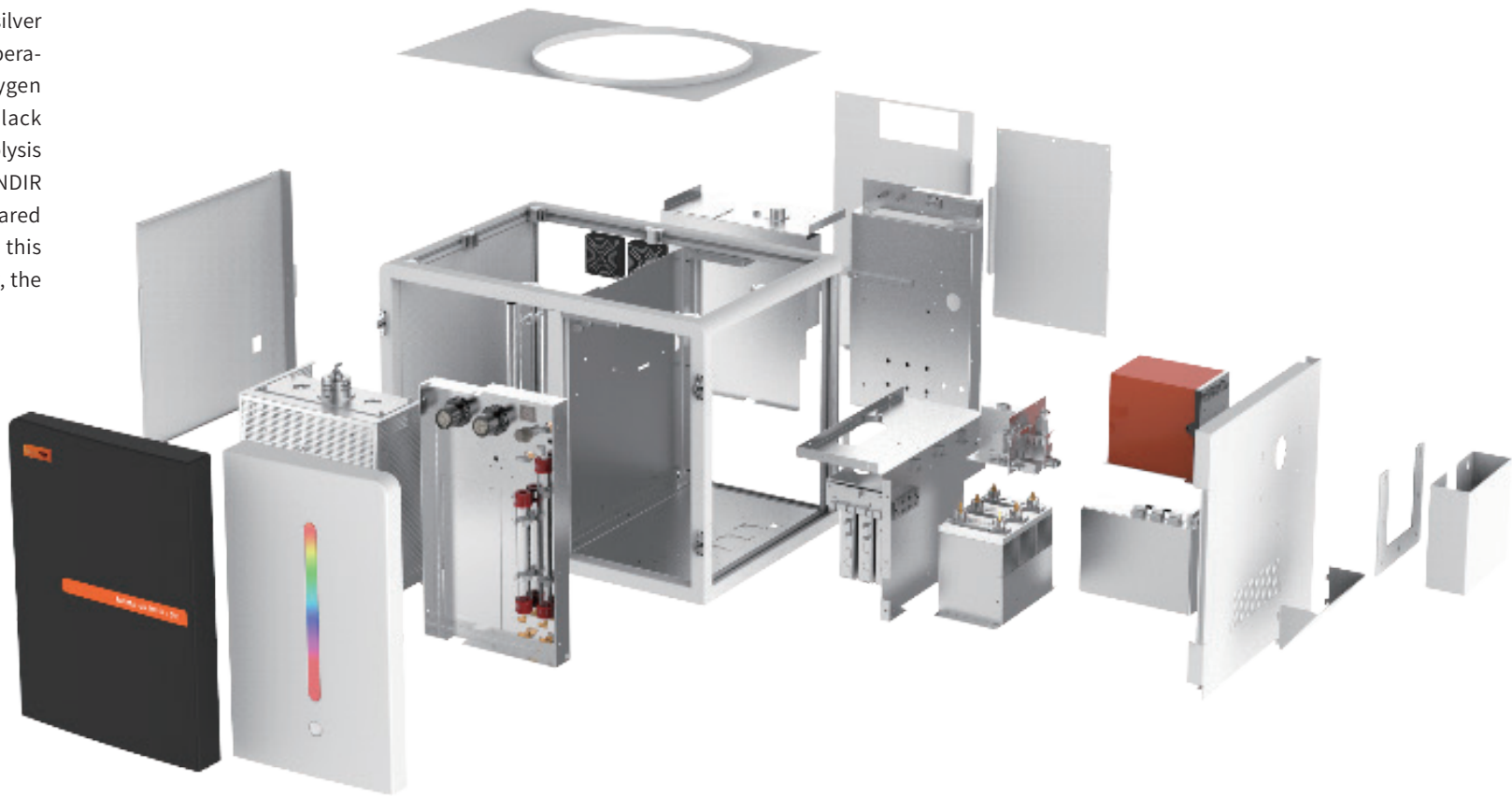
Oxygen Mode: Oxygen-containing organic samples are wrapped in silver boats/silver capsules and delivered by a crucible into the high-temperature pyrolysis tube. Under an inert gas atmosphere at 1150°C, oxygen radicals formed during pyrolysis react with high-purity carbon black packed within the pyrolysis tube to generate CO. Other gaseous pyrolysis products, along with the CO, are carried by the carrier gas into an NDIR (Non-Dispersive Infrared) detector. The absorption intensity of infrared light at the characteristic wavelength of CO is measured. Based on this absorption, the CO concentration is determined, and subsequently, the oxygen content is calculated.



The E500 Elemental Analyzer is a high-end, precision laboratory instrument for the analysis of major elements in organic matter. It enables rapid and accurate quantitative analysis of C, H, N, S, and O elements in solid and liquid samples within a single instrument, playing a crucial role in studying the elemental composition of organic materials and compounds. It finds wide application in scientific research, agriculture, food, petrochemical/chemical industries, geology/mining, pharmaceuticals, and other fields.

Measuring the content of carbon, hydrogen, oxygen, nitrogen, and sulfur in samples reveals the composition, properties, and changes of compounds, serving as an effective tool for scientific research.

This product is the result of Hanon Technology Group Co., Ltd.'s research and development project undertaken as part of Shandong Province's Major Scientific and Technological Innovation Project. It has successfully passed the project acceptance review by the Shandong Provincial Department of Science and Technology and received high praise from industry experts.





Instrument Features

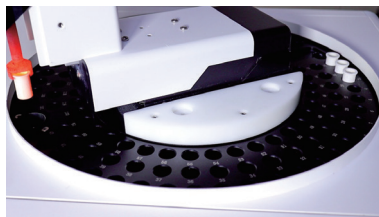
1、In-situ High-Temperature Sample-Ash Replacement

Technology: Eliminates the need to manually clean residual ash from the combustion tube, significantly improving instrument efficiency and detection accuracy.

Explanation: Samples are contained within crucibles placed inside the combustion/pyrolysis tube. This solves the problems of frequent instrument downtime for manual ash removal and the inability to continuously analyze large batches of samples in traditional instruments. Simultaneously, it prevents interference from residual ash on subsequent samples, ensures complete sample combustion, and thereby significantly enhances detection efficiency and accuracy.

The autosampler features a 120-position rotary sample tray for convenient loading, with crucible arrangement being clear and intuitive.

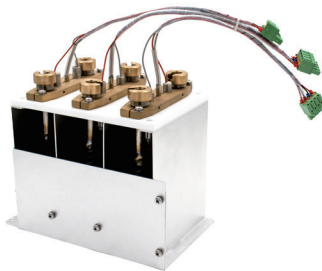
Ceramic crucibles are high-temperature resistant, oxidation resistant, high-strength, contamination-free, and reusable after cleaning post-test.



2、High-Temperature Aluminosilicate Fiber Combustion Furnace: Utilizes coupling control technology for the temperature field and carrier gas flow field during combustion/pyrolysis, significantly enhancing working temperature, and the uniformity and stability of the temperature field. Maximum working temperature reaches 1400°C. Length of stable temperature zone $\geq 200\text{mm}$. Temperature uniformity deviation within the stable zone length $\leq \pm 10^\circ\text{C}$. Achieves effective control over the uniformity and stability of the temperature field.



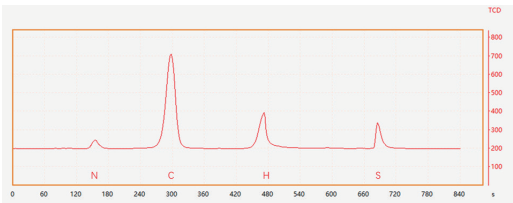
3、Specific Adsorption-Desorption Technology: Developed to address the technical challenge of low detection accuracy for trace elements, significantly enhancing the instrument's detection precision.



Based on the properties of organic element combustion products CO_2 , H_2O , and SO_2 , specific molecular sieve materials for targeted adsorption-desorption were developed. Three adsorption columns specifically adsorb H_2O , SO_2 , and CO_2 respectively through physical adsorption.

After the analysis of one element is complete, the next gas component is released via programmed heating for detection, enabling efficient adsorption and desorption of the target gas mixture. Adsorption is complete, desorption is rapid, gas separation is highly efficient and thorough, effectively avoiding peak tailing issues common in chromatographic separation methods.

Furthermore, physical adsorption ensures excellent separation reproducibility and long column lifespan.



4、Newly Designed High-Flow, High-Sensitivity TCD: Incorporates technologies such as constant power filament control, full digital calibration, digital filament balance, and digital-analog signal isolation. This reduces baseline noise while avoiding temperature fluctuation effects, effectively improving the sensitivity and quantitative repeatability of dynamic gas component measurement. Oxidation-resistant filaments ensure no damage even if trace oxygen is present.

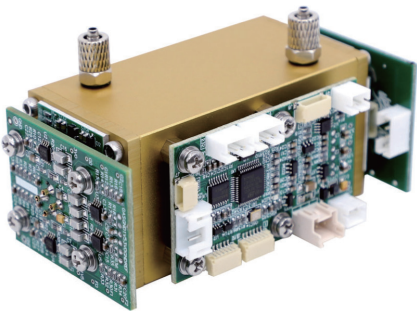
Electronic Mass Flow Controllers (MFC) precisely control the stability and consistency of carrier gas flow and pressure.

Drying and purification tubes provide a clean environment for the detection process.



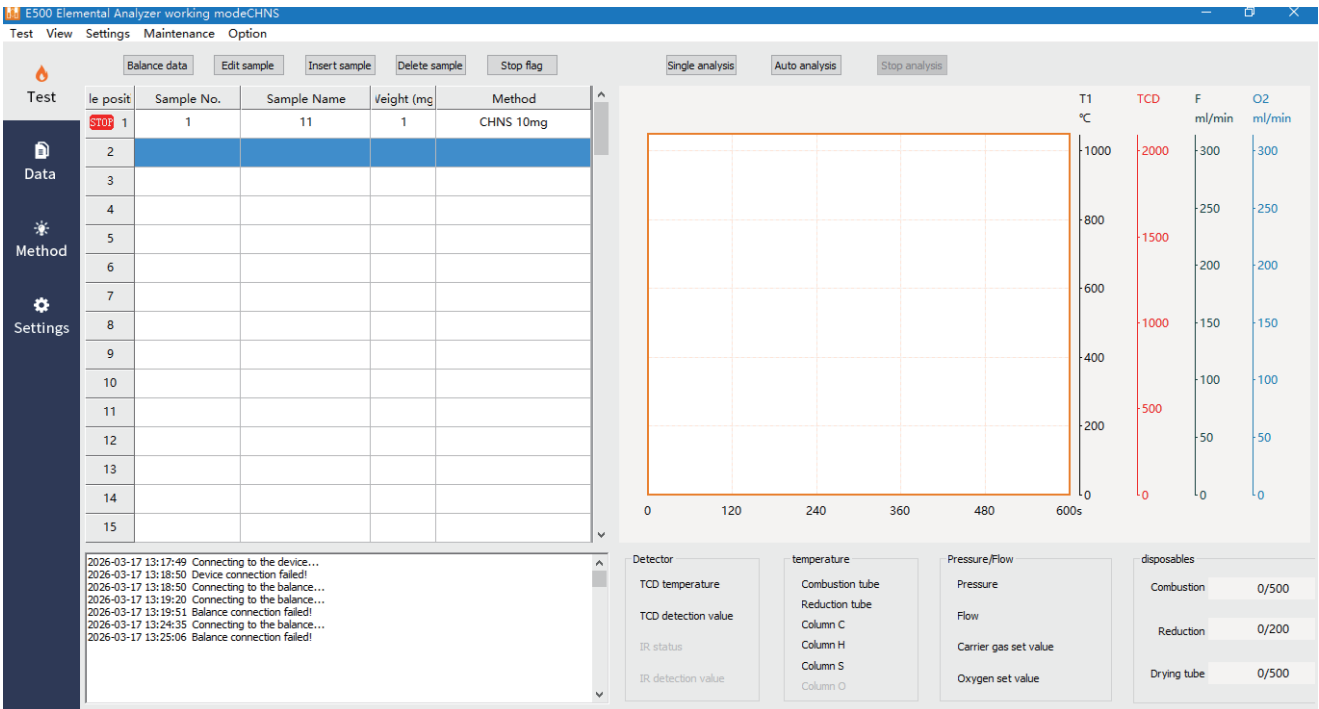
5、Oxygen Mode - NDIR Detector: Employs a highly specific Non-Dispersive Infrared (NDIR) detector. At the wavelengths of $4.67\mu\text{m}$ and $4.72\mu\text{m}$, CO gas molecules exhibit selective absorption (other pyrolysis gas products do not absorb). The absorption value has a linear relationship with CO concentration (following the Lambert-Beer law). The CO concentration in the sample is determined based on the infrared light absorption value, from which the oxygen content is derived.

The NDIR detector features low detection limits and strong anti-interference capability.





SoftwareWorkstation

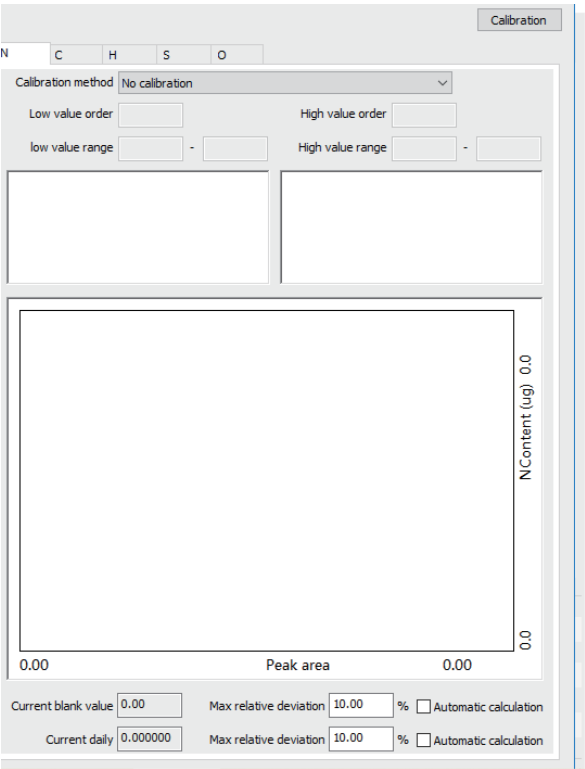


Self-developed software workstation, fully functional and user-friendly.

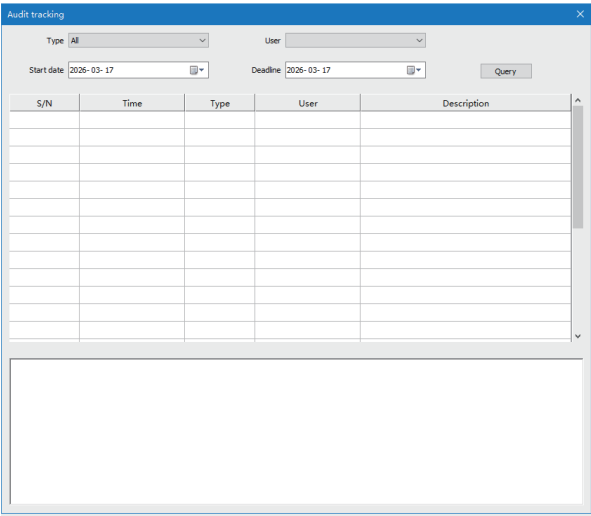
User-Friendly Interface, Simple and Easy to Use:

- Flat design interface for quick completion of sample testing, data processing, method editing, and instrument configuration.
- Top of the testing interface provides commonly used function areas, enabling rapid sample testing operations without multi-level menus.
- Lower right area of the testing interface displays real-time instrument status and working parameters.

Software Functions



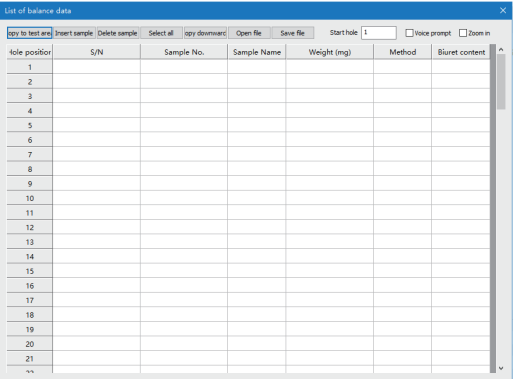
—Calibration Curve Method:Linear or non-linear calibration; users can customize the calibration curve order.



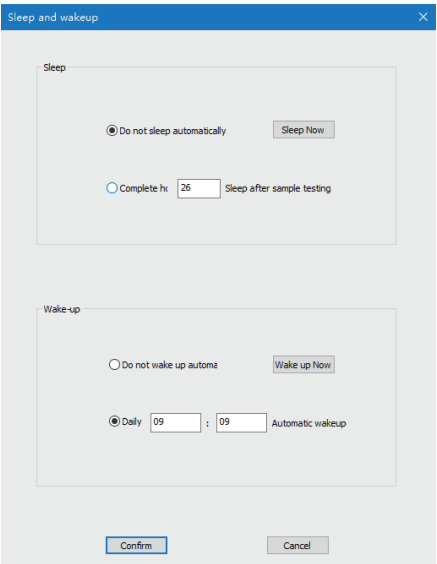
—Audit Trail:Software design complies with FDA 21 CFR Part 11 requirements. User permissions can be set with 3 levels of management; unlimited number of new users can be added. All instrument operations and setting changes are recorded and traceable, ensuring more authentic and secure experimental data.



Practical Features



—**Wireless Transmission of Balance Weighing Data:**Utilizes industrial-grade wireless transmission modules to automatically collect balance weighing data, reducing workload and avoiding errors.



—**Standby/Wake-up Function:**After analysis is complete, the instrument can automatically cool down and enter sleep/standby mode, and can be scheduled to wake up automatically. This reduces the frequent wait times associated with heating/cooling during startup/shutdown, and the standby state also saves carrier gas and reduces instrument power consumption.



—**Delayed Air Cooling:**After shutdown post-analysis, the cooling fan continues to operate until the furnace temperature drops to a safe level before automatically turning off. This saves cooling wait time and extends equipment lifespan.

Application Fields

Chemical/Pharmaceutical:
Fine chemicals, new material development, new drug R&D, etc.

Agriculture and Environment:
Plants, food, soil, mixed fertilizers, wood, etc.

Environmental Monitoring:
Wastewater, solid waste, sludge/sediments, etc.

Geology/Mining/Petrochemical:
Rocks and minerals, coal/coke, petrochemical products, fuels/-fossil fuels, etc.

Sample Test Data

Main Consumables:
High-purity, high-quality consumables are selected to ensure precise and reliable analysis results.

Name	Specification
Ceramic Crucible	95% Porcelain, LBJ-E500-005
Oxygen Lance	Corundum Tube, LBJ-E500-373
Quartz Combustion Tube	Quartz, LBJ-E500-003
Quartz Reduction Tube	Quartz, LBJ-E500-092
Drying Tube Assembly	ZPT-E500-032
Quartz Support Tube	Quartz, LBJ-E500-375
Quartz Support Rod	Quartz, LBJ-E500-374
High-Purity Alumina Balls	50g / Bottle
Wire Copper	100g / Bottle
High-Purity Quartz Wool	100g / Box



Name	Specification
Tungsten Trioxide (WO ₃)	60g / Bottle
Benzoic Acid	5g / Bottle
Silver Wire	50g / Pack
Desiccant	454g / Bottle
Graphite Felt	φ25*10mm, 10pcs/Pack
Carbon Black	80g / Bottle
Sulfanilamide	10g / Bottle
Tin Foil	400pcs/Box, 35*35mm
Tin Boats	500pcs/Pack, 6*6*12mm
Silver Boats	114pcs/Box, 6*6*12mm
Sample Box	24-Well Sample Box, PS, D100

Technical Specifications

Autosampler	120-position sample tray, crucible introduction, in-situ high-temperature replacement technology.
Gas Separation Method	Adsorption-Desorption separation technology.
High-Temperature Combustion & Reduction	Combustion temperature up to 1400°C. Ensures complete sample combustion.
Analysis Time	Approx. 3-4 min per element*.
Sample Size	Maximum 1.5g or sample volume 1 mL.
Detection Range	Maximum dynamic range for element concentration:
	C: 0-30mg absolute or 0 - 100%
	H: 0-4mg absolute or 0 - 100%
	N: 0-10mg absolute or 0 - 100%
	S: 0-5mg absolute or 0 - 100%
	O: 0-3mg absolute or 0 - 100%
Repeatability Error	Standard Deviation ≤ 0.1% (10mg Sulfadiazine standard).
Detection Limit	C, H, N, S: ≤ 30ppm (TCD detector)
	O: ≤ 20ppm (IR detector)
Detectors	C/H/N/S Mode: Thermal Conductivity Detector (TCD)
	O Mode: Non-Dispersive Infrared Detector (NDIR)
Working Gases	Carrier Gas: Helium (He), Purity: 99.999%
	Combustion Gas: Oxygen (O ₂), Purity: 99.999%
Dimensions (L×W×H)	735mm × 560mm × 1160mm (including autosampler)
Net Weight	100kg (including autosampler)

Depends on sample type, analysis mode, and parameter configuration. Optimized automatically based on element content and sample weight.

Operating Conditions

Power Supply	220V AC ±10%, 50Hz
Computer	PC with Windows 7 or later operating system, equipped with USB or RS 232 interface.
Temperature	Operating Environment 15°C – 30°C
Humidity	≤ 85% RH